MILITARY SPECIFICATION

CLOTH, DOUBLE - PLAIN WEAVE, COTTON, WHITE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers the requirements for one type of cotton cloth.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

PPP-P-1134 - Packaging and Packing of Cotton and Cotton Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8305

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2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

RULES AND REGULATIONS UNDER THE TEXTILE FIBER PRODUCTS IDENTIFICATION ACT

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D.C. 20580.)

3. REQUIREMENTS

3.1 Standard sample.- The cloth shall match the standard sample for shade, and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article approval.- The requirement for the first article will be as specified by the procuring activity (see 6.2).

3.3 Cloth, cotton, white.- The cloth shall be of combed cotton which has been spun into 2-ply yarn for the warp and either single or 2-ply yarn for the filling. The finished cloth shall conform to the requirements shown in Table I when tested as specified in 4.4.

Table I - Physical requirements

Weight per square yard (min.)	Yarns per inch (min.)		Breaking strength pounds (min.)		Total shrinkage percent (max.)	
Ounces	Warp	Filling	Warp	Filling	Warp	Filling
9.0	210	108	125	125	2.0	1.0

3.4 Weave.- The cloth shall be a double fabric consisting of two warps bound together by a single filling and a binder warp. The face shall weave plain with picks showing on the face. The back shall weave plain with picks showing. The binder warp shall weave plain or may weave two up-two down, every other end opposite. There shall be four ground ends between each binder warp end. The filling shall weave alternately on the face and on the back.

3.5 Width.- The width of the cloth shall be as specified (see 6.2) and shall be the minimum width inclusive of the selvage.

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3.6 Color.- The color of the cloth shall be white, shade No. 3035.

3.6.1 Matching.- The finished cloth shall match the standard sample under natural (north sky) daylight or artificial daylight, having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.7 Finish.- The cloth shall be singed, desized, fully mercerized and preshrunk. The preshrinking process shall not be identified by name or trademark on either the ticket, the cloth or the package. The cloth shall be fully bleached and tinted to match the color specified (see 3.6).

3.8 pH.- The pH value of the water extract of the finished cloth shall not be less than 5.0 nor more than 8.5 when tested as specified in 4.4.

3.9 Length and put-up.- Unless otherwise specified in the contract or purchase order, the cloth shall be furnished in pieces of not less than 50 continuous yards in length and shall be furnished on rolls as specified in PPP-P-1134.

3.10 Fiber identification.- Each roll of cloth shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.11 Marking.- The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "Face" at each end of the piece.

3.12 Workmanship.- The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards when calculated as specified in Section 4, shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Certificate of compliance.- Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

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4.2 First article inspection.- When required, the first article, submitted in accordance with 3.2, shall be visually inspected for color, finish and appearance and shall be tested for chemical and physical properties in accordance with the applicable methods specified in 4.4.

4.3 Inspection.- Inspection shall be in accordance with MIL-STD-105, except where otherwise indicated herein.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.2 Examination of the end item.- Examination of the end item shall be in accordance with 4.3.2.1 through 4.3.2.5.

4.3.2.1 Yard-by-yard examination.- The entire yardage of each roll in the sample, but no more than 100 yards per roll shall be examined on the face side. Defects as defined in Section 1 of FED-STD-4 which are clearly visible at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.3.2.2, except that only those slubs and knots which exceed the limits shown on Sears Fabric Defect Scales (see 6.4), C for slubs and B for knots, shall be scored. No continuous linear yard from any one roll within the sample shall be penalized more than 4 points. The sample size shall be twenty rolls selected from twenty containers. For determination of acceptability on the average quality of the lot, the lot shall be unacceptable if the points per 100 square yards exceed 30 points. For determination of acceptability on the individual roll quality basis, the lot shall be unacceptable if two or more individual rolls exceed 45 points. If no individual roll exceeds the point level, the lot shall be accepted. If one roll exceeds the point level, a second sample of 20 rolls shall be selected for individual roll quality examination only. If no rolls in the second sample exceeds the point value, the lot shall be accepted. If one or more rolls exceed the value, the lot shall be rejected. For individual roll quality examination, only local type defects shall be scored. Overall type defects shall be scored only in determining the lot average quality requirement. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{Total yards inspected}} = \text{Points per 100 square yards}$$

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4.3.2.2 Demerit points.- Demerit points shall be assigned as follows:

For defects up to and including 3 inches in any dimension	- one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

The following defects, when present, shall be scored 4 points for each yard in which they occur:

Selvage curled, slack, tight, corded, cut or torn
Bias or bowed filling
Baggy, ridgy or wavy cloth
Overall uncleanness
Width not as specified

4.3.2.3 Examination for length.-

4.3.2.3.1 Individual pieces.- During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified, or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.3.2.3.2 Total yardage in sample.- The lot shall be unacceptable, if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.

4.3.2.4 Examination for shade.- During the yard-by-yard examination, each roll in the sample shall be examined for shade. Any roll in the sample off-shade, shaded side to side, side to center or end to end shall be cause for rejection of the entire lot represented by the sample.

4.3.2.5 Examination for face markings and for compliance with the Textile Fiber Products Identification Act.- During the yard-by-yard examination, each roll in the sample shall be examined for face markings and for compliance to the Textile Fiber Products Identification Act. The lot shall be unacceptable if two or more pieces in the sample have the face stamping missing from either or both ends, or the face stamping on the wrong side; or are not labeled, ticketed, or invoiced in accordance with the Textile Fiber Products Identification Act.

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4.3.3 Examination of preparation for delivery requirements.- An examination shall be made in accordance with the provisions of PPP-P-1134 to determine that packaging, packing and marking requirements of Section 5 of this specification are complied with.

4.4 Testing of the end item.- The methods of testing specified in FED-STD-191, wherever applicable, and as listed in Table II shall be followed. The physical and chemical values specified in Section 3 apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test method. The sample unit shall be 3 1/2 linear yards, full width, of the finished cloth. The sample size shall be as shown below. The lot size shall be expressed in units of 1 yard each. The lot shall be unacceptable if one or more units of product fail to meet any of the requirements specified.

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and up	5

Table II - Test methods

Characteristic	Requirement reference	Test method
Fiber content	3.3	1/
Weight	Table I	5041
Yarns per inch:		
Warp	Table I	5050
Filling	Table I	5050
Breaking strength:		
Warp	Table I	5100
Filling	Table I	5100
Shrinkage:		
Warp	Table I	5550
Filling	Table I	5550
Weave	3.4	Visual 2/
Finish	3.7	1/
pH	3.8	2811

1/ A Certificate of Compliance will be acceptable for this requirement.
2/ One determination per sample unit and the result reported as "Pass" or "Fail".

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C, as specified (see 6.2 and 6.2.1).

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5.1.1 Levels A and C.- The cloth put up as specified (see 3.7), shall be packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C.- The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking.- In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1134.

6. NOTES

6.1 Intended use.- The cloth will be utilized to manufacture cap crowns used by men and women Navy personnel.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Requirements for the First Article (see 3.2).
(c) Width required (see 3.5).
(d) Levels of packaging and packing required (see 5.1 and 5.2).

6.2.1 In the preparation of contracts or purchase orders, it must be noted that various options, choices, and alternatives, as indicated in PPP-P-1134 may be exercised in the preparation for delivery of the cloth.

6.3 Standard sample.- For access to the standard sample, address the procuring activity issuing the invitation for bids.

6.4 Sears Fabric Defects Scales are available from: Sears Roebuck and Co.,; Dept. 769; "Fabric Defect Replica Kit"; 925 South Homan Avenue; Chicago, Illinois 60607.

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